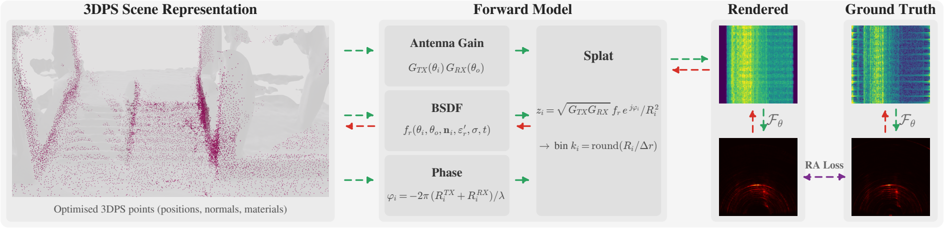

3DPS Scene Representation
Optimised 3DPS points (positions, normals, materials)
Forward Model
Antenna Gain
$G_{TX}(\theta_i)\,G_{RX}(\theta_o)$
BSDF
$f_r(\theta_i, \theta_o, \mathbf{n}_i, \varepsilon_r', \sigma, t)$
Phase
$\varphi_i = -2\pi\,(R_i^{TX} + R_i^{RX})/\lambda$
Splat
$z_i = \sqrt{G_{TX}G_{RX}}\; f_r\, e^{j\varphi_i}/R_i^2$
$\to$ bin $k_i = \mathrm{round}(R_i/\Delta r)$
Rendered
Ground Truth
$\mathcal{F}_\theta$
$\mathcal{F}_\theta$
RA Loss